OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM
SYSTEM:	Residual Heat Removal
TASK:	TCAF Loss of RHR cooling
TASK NUMBER:	N1140300401
JPM NUMBER:	14-01 NRC Sim h

ALTERNATE PATH: ☐

K/A NUMBER: APE025 AA1.01/AA1.03

IMPORTANCE FACTOR: 3.6/3.4 (RO) 3.7/3.3 (SRO)

APPLICABILITY:
EO ☐ **RO** ☒ **STA** ☐ **SRO** ☒

EVALUATION SETTING/METHOD: Simulator

REFERENCES: S2.OP-AB.RHR-0001, Rev. 18 Loss of RHR (rev. checked 9-28-15)

TOOLS AND EQUIPMENT: None

VALIDATED JPM COMPLETION TIME: 8 min

TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS: N/A

Developed By:	**G Gauding** **Instructor**	**Date: 4-24-15**
Validated By:	**S Bickhart / M Spencer** **SME or Instructor**	**Date: 9-28-15**
Approved By:	DL Williams **Training Department**	**Date:** 10-15-15
Approved By:	(MOLS) **Operations Department**	**Date:** 10 13 15

ACTUAL JPM COMPLETION TIME:

ACTUAL TIME CRITICAL COMPLETION TIME:

PERFORMED BY: ______

GRADE: ☐ SAT ☐ UNSAT

REASON, IF UNSATISFACTORY:

EVALUATOR'S SIGNATURE: **DATE:**

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SIMULATOR SETUP INSTRUCTIONS

SYSTEM:	Residual Heat Removal
TASK:	TCAF Loss of all RHR cooling
TASK NUMBER:	N1140300401
SIMULATOR IC:	**14-01 NRC JPM IC-258** Shutdown IC with SI pumps, Accumulators, and one centrifugal charging pump removed from service, 22 RHR pp cont pwr off
MALFUNCTIONS REQUIRED:	**RT-1 RH0026A 21 RHR pump trip**
OVERRIDES REQUIRED:	NONE
SPECIAL INSTRUCTIONS:	Place RHR system on P-250 computer. Place 21 RHR HX inlet temp on trend and T0002A CET temp with 400°F high setpoint.

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NAME: ____________________

DATE: ____________________

SYSTEM: Residual Heat Removal

TASK: TCAF Loss of all RHR cooling

TASK NUMBER: N1140300401

INITIAL CONDITIONS:

- Reactor is shutdown.
- 21 RHR loop is in service for shutdown cooling.
- 22 RHR loop is aligned for ECCS.
- 22 RHR pump breaker is racked down to check condition after scaffolding construction heavily impacted cubicle door and caused actuation of OC trip relay.
- 21 RHR HX inlet temp is 329°F and stable.
- RCS pressure is 321 psig and stable.
- 22 and 23 RCPs are in service.
- 21 and 22 AFW pumps in service while filling SGs.
- 22 and 23 Charging pumps and both SI pumps are C/T.
- The accumulators have been isolated.
- There are no personnel in containment.

INITIATING CUE:

Maintain current conditions.

Successful Completion Criteria:

1. **All critical steps completed.**
2. **All sequential steps completed in order.**
3. **All time-critical steps completed within allotted time.**
4. **JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).**

Task Standard for Successful Completion:

1. Feed SG(s) and throttle open MS10(s) as required to establish CET temperatures stable or lowering.

NAME: ____________________

DATE: ____________________

SYSTEM: Residual Heat Removal

TASK: TCAF A Loss of All RHR

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
		INSERT RT-1 after operator assumes the watch.			
			Announces unexpected 21 RHR pump trip.		
			Enters S2.OP-AB.RHR-0001, Loss of RHR.		
	3.1	**INITIATE** Attachment 1, Continuous Action Summary.	Initiates Attachment 1, Continuous Action Summary.		
	3.2	IF the RCS is vented to the Containment atmosphere with the Containment Equipment hatch OPEN AND at least two RCS loops are filled with associated SG's available, THEN **CLOSE** the vent path prior to Core Boil. (Refer to Attachment 4)	Determines RCS is not vented to containment.		
*	3.3	Is RCS aligned for operation <101 ft. elevation (Reduced Inventory)?	Determines RCS is not aligned for operation <101 ft. elevation (Reduced Inventory)		

OPERATOR TRAINING PROGRAM

JOB PERFORMANCE MEASURE

NAME: ____________________

DATE: ____________________

SYSTEM: Residual Heat Removal

TASK: TCAF A Loss of All RHR

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	3.5	Is the loss of RHR due to a mechanical failure or loss of electrical power to the in-service RHR Pump?	**Cue:** "Maintenance reports that the loss of 21 RHR pump is due to a loss of electrical power to pump." Goes to Step 3.50.		
*	3.50	Is a heat sink available for Residual Heat Removal? • Component Cooling to RHR System • Service Water to Component Cooling System	Determines a heat sink is available for Residual Heat Removal		
*	3.51	Is an RHR loop available?	Determines 22 RHR loop is not available, and no RHR loop is available, goes to step 3.31.		
	3.31	CONTINUE			

OPERATOR TRAINING PROGRAM

JOB PERFORMANCE MEASURE

NAME: ______________________

DATE: ______________________

SYSTEM: Residual Heat Removal

TASK: TCAF A Loss of All RHR

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	3.32	**INITIATE** <u>one</u> of the alternate methods of decay heat removal: • Attachment 7, Hot Leg Injection (Feed & Bleed - Preferred method if RCS NOT intact or Loops NOT filled AND core exit TCs $\geq$200°F) • Attachment 8, Cold Leg Injection (Feed & Bleed - Preferred Method if Core Exit TCs < 200°F) • Attachment 9, Steam Generator Reflux Cooling (RCS depressurized AND no other means of decay heat removal is available) • Attachment 10, Forced Flow Or Natural Circulation Cooldown (RCS intact and filled to greater than 0% in the Pressurizer with Loops filled) • Attachment 11, Cooling the RCS with Spent Fuel Pool (Reactor Vessel Head Removed)	Determines correct attachment is Attachment 10 based on: RCS is intact, filled, and pressurized. CET's are >200°F		

NAME: ____________________

DATE: ____________________

SYSTEM: Residual Heat Removal

TASK: TCAF A Loss of All RHR

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	Att. 10 1.0	**FEED** available Steam Generators to maintain wide range level >77% using Auxiliary Feedwater System or Condensate System.	Feeds available Steam Generators to maintain wide range level >77% using Auxiliary Feedwater System. (All Steam generators are available) **Cue if Required: If rate of feed flow will result in a long fill time to reach 77% WR level, then cue that all SG WR levels are >77%.**		
*	Att. 10 2.0	**REMOVE** reactor decay heat by performing one of the following: **OPERATE** the appropriate MS10s to maintain Core Exit Thermocouples stable or lowering. OR **DRAIN** Steam Generators as required to maintain level <95% wide range.	Removes reactor decay heat by operating the appropriate MS10s to maintain Core Exit Thermocouples stable or lowering. OR Determines that the rate of fill of SGs is currently maintaining Core Exit Thermocouples stable or lowering.		

TERMINATING CUE: After determination that CETs are stable or dropping, terminate JPM.

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

________ 1. Task description and number, JPM description and number are identified.

________ 2. Knowledge and Abilities (K/A) references are included.

________ 3. Performance location specified. (in-plant, control room, or simulator)

________ 4. Initial setup conditions are identified.

________ 5. Initiating and terminating Cues are properly identified.

________ 6. Task standards identified and verified by SME review.

________ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).

________ 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 18 Date S2.OP-SO AB.RHR-0001

________ 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.

________ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.

________ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

Michael Spector

SME/Instructor: Michael Spector Date: 9/28/15

SME/Instructor: Burkhart Date: 9/28/15

SME/Instructor: ________________ Date: ________

INITIAL CONDITIONS:

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INITIATING CUE:

Maintain current conditions.